# Is the galaxy RAR scale $cH_0/6$? A data-first reading of 12 X-COP cluster profiles and 13 CLASH cores

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**Student version.** This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23506655

## Abstract

The radial acceleration relation (RAR) of disc galaxies has a single scale, $a_0 = (1.20 \pm 0.02 \pm 0.24)\cdot 10^{-10}$ m s$^{-2}$, numerically close to $cH_0$ (Milgrom: $cH_0/2\pi$; Verlinde's emergent-gravity construction: $cH_0/6$). Galaxy clusters show a similar low-acceleration relation with scales about ten times higher. This record looks at three measured things and then reads them with the author's framework, Mathematical Materiality (MM). The author's method is to go from the data to the framework: readings found after seeing the data are stated as such, protocols and local registrations are kept, and the checks by a second AI assistant are documented. (1) The galaxy scale. MM's two-end reading gives $cH_0/2$, 2.73 times the measured $a_0$. Dividing by 3 gives $a_0 = cH_0/6 = 1.09$ ($H_0 = 67.4$) or 1.18 ($H_0 = 73.0$) $\cdot 10^{-10}$ m s$^{-2}$, inside the measured band; in MM the factor is read as the rule used in MM 2.44 for Ω_c/Ω_b = 3√3, counted on two directions. The count, the direction of the factor and the identification with $a_0$ were chosen knowing the target. Approximate coherence checks: vertical density near the Sun (63.9 vs 68 ± 4 M⊙ pc$^{-2}$, against 107 with $cH_0/2$), the 3√3 cluster baryon fraction (unchanged), the shape of $a_0 \propto H(z)$ (compatible with free amplitude if the MUSE-DARK III bars are 1σ).

# Is the galaxy scale $cH_0/6$? A data-first reading of 12 X-COP clusters and 13 CLASH cores

*Student version of MM record 2.46.1 (DOI 10.5281/zenodo.23506655), written with an AI assistant from the archived text; the archived text remains on Zenodo.*

The archived text is in English.

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**How to read this text.** First we explain the words you need. Then comes a summary, *In short*. Sections 1 to 8 go into the details. At the end you find how the work was done and a glossary. Numbers like 2.44 or 2.46 indicate other works of the author, archived on Zenodo.

## Before starting: the words you need

**Galaxies and clusters.** A *galaxy* is a huge collection of stars, gas and dust held together by gravity. *Disc* galaxies are flat and rotate around their centre. A *galaxy cluster* is a very large group of galaxies bound by gravity, with a very hot gas between them. A *group* is like a cluster, but smaller.

**Acceleration.** It says how much the velocity changes every second, in size or in direction. It is measured in m s$^{-2}$, metres per second squared. A star that orbits the centre of a galaxy has an acceleration towards the centre.

**The missing mass.** From the motion of stars and gas one derives how much mass there is. In galaxies and clusters this mass is larger than the mass we see. The matter we see, made of atoms, is called *baryonic*. There are two kinds of answer: an invisible matter, *dark matter*; or a different gravity when the acceleration is very small, as in MOND, the idea of the physicist Milgrom.

**The RAR and its scale $a_0$.** In disc galaxies an observed law holds, the *radial acceleration relation*, RAR for short. It links two accelerations:

- g_obs, the observed one, derived from the motion;
- g_bar, the one that baryonic matter alone would give.

The formula is g_obs = g_bar/(1 − e^(−√(g_bar/$a_0$))). The law has a single *scale*, $a_0$ (read *a-zero*). In words: when g_bar is much larger than $a_0$, g_obs is almost equal to g_bar: Newton holds. The more g_bar goes down towards $a_0$ and below, the more g_obs exceeds g_bar: this is the *dark part*, that is, the extra gravity. The passage is gradual; there is no sharp point.

**$cH_0$.** c is the speed of light. $H_0$ (read *H-zero*) is the *Hubble constant*: it says how fast the universe is expanding today. The product $cH_0$ is an acceleration.

**The dex.** It measures ratios in powers of ten. 1 dex means ten times. Zero dex means equal. An example with numbers chosen by us: 0.1 dex means about 26 % more.

**The radii of a cluster.** R500 is the radius within which the mean density of the cluster is 500 times the *critical density*, a reference density of the universe. R200c is the same with 200 times. R200m uses 200 times the mean density of matter in the universe.

**The splashback radius.** It is the point where the matter falling into a cluster turns back for the first time: a sort of natural edge of the cluster.

**Hydrostatic and lensing masses.** The hot gas of a cluster is in *hydrostatic equilibrium*: its pressure balances gravity. From the temperature and the density of the gas one derives the total mass: it is the *hydrostatic mass*. Another way of weighing is *gravitational lensing*: the mass bends the light of the galaxies that lie behind.

**NFW.** It is a mass profile typical of dark-matter haloes in computer simulations.

**Two statistical measures.**

- The *Spearman coefficient* is a number between −1 and +1. It says how much two quantities grow together. Close to zero means no link.
- The $\chi^2$ (chi-squared) measures how far a model departs from the data. The *degrees of freedom* count the independent comparisons. The *p-value* is the probability of finding a deviation this large by pure chance.

**The words of the MM framework.** Mathematical Materiality, MM for short, is the author's framework of ideas.

- A *premise* is a starting sentence of the author: a fixed point, like a postulate. The premises are numbered.
- A *reading* is an interpretation, written so that it can be put to the test.
- *Registered* means written and fixed with its date before doing the calculations.
- The *hidden sector*, or second *mirror*, is an idea of MM 2.44: a matter that we do not see, read as dark matter, next to the visible sector.
- The *distal end* is the far end of a relation.

## In short

**The question.** The RAR of disc galaxies has a single scale:

$a_0 = (1.20 \pm 0.02 \pm 0.24) \cdot 10^{-10}$ m s$^{-2}$

- $10^{-10}$ means divided by ten billion. The two uncertainties are the statistical one (± 0.02) and the systematic one (± 0.24).
- This value is numerically close to $cH_0$. Milgrom proposes $cH_0/2\pi$. Verlinde's *emergent gravity* construction gives $cH_0/6$.
- In galaxy clusters a similar relation holds at low acceleration, but with scales about ten times higher.

The record looks at three measured things. Then it reads them with the MM framework.

**1. The galaxy scale.**

- MM's *two-end reading* gives $cH_0/2 = 3.27 \cdot 10^{-10}$ m s$^{-2}$. It is 2.73 times the measured $a_0$.
- Dividing by 3 one gets $a_0 = cH_0/6$. It is $1.09 \cdot 10^{-10}$ m s$^{-2}$ with $H_0 = 67.4$, or $1.18 \cdot 10^{-10}$ with $H_0 = 73.0$. Both

values lie inside the measured band (section 3).
- In MM the factor 3 is read as the rule that MM 2.44 uses for the ratio Ω_c/Ω_b = 3√3: a factor √3 for each direction between the visible sector and the hidden one. Here it is counted on two directions: √3 times √3 makes 3.
- **Be careful:** the count, the direction of the factor and the identification of the threshold with $a_0$ are choices made knowing the target.

**2. Twelve X-COP clusters.** X-COP is a sample of clusters studied in depth with X-ray satellites.

- The scale is derived point by point, with the galaxy formula. In all twelve clusters the scale decreases with radius. The median slope is −0.97, with a range from −1.44 to −0.61, between 0.2 and 1.2 R500.
- A scale A/r, with a different *amplitude* A for each cluster, describes every profile better than a constant scale.
- A registered test looked for a dependence on the "history" of the cluster: one scale per cluster at R500, compared with the *concentration* and with the *central entropy*. It gives no indication (Spearman −0.09 and +0.02).
- In MM the A/r behaviour is read as a *fixed distal end*: the quantity that counts becomes G·M_bar(<r)/r.
- An ordinary explanation gives the same slope: an NFW-type total mass, with a baryon fraction that grows outwards (section 5).

**3. Thirteen CLASH cores.** CLASH is a large programme of observations of clusters. The *core* is the central region, inside the brightest galaxy of the cluster (14.3-30.3 kpc).

- The registered test gives no indication. In 10 cores out of 13 the baryonic mass written in the tables exceeds the lensing mass.
- In an *exploratory* test, the two cluster laws of point 2, extended inwards, predict from 2 to 15 times the lensing mass (section 6).

**4. Readings that fall.** A fourth part records readings that fall.

- Written with counts, the onset of the dark part is at c⁴/(4G·M_U), with the *capacity rule* and spherical-equivalent masses.
- Eight readings tie it to the centre of each galaxy. They give scales that change from galaxy to galaxy, while the RAR has a single scale. These readings fall (section 4).

**What version 2.46.1 adds.** A new check shows that the radius of the fixed end depends strongly on the mass profiles adopted. If the total masses grow by 10 %, the radius grows by about 23 %. So its closeness to the splashback radii depends on the calibration of the masses (section 5).

**The method, declared.** The author's instruction is to go from the data to MM. A reading found after seeing the data is acceptable: the criterion is *coherence*. The readings built knowing the target and the analyses added afterwards are marked as such.

## 1. What a physicist will recognise

This section lists the things already known, from which the record starts.

- **The RAR** of McGaugh, Lelli and Schombert (2016): g_obs = g_bar/(1 − e^(−√(g_bar/$a_0$))). The closeness of $a_0$ to $cH_0$ is also known:
- Milgrom: $a_0 \approx cH_0/2\pi$;
- Verlinde (2016): a_M = $cH_0/6$, with a law that depends on the profile for extended systems.
- **The RAR in clusters.** The scales found are:
- CLASH $2.02\cdot10^{-9}$ (Tian et al. 2020). *Et al.* means *and others*;
- X-COP $1.12\cdot10^{-9}$ (Pradyumna et al. 2021);
- groups observed in X-rays: $0.88\cdot10^{-9}$ m s$^{-2}$ (Gopika and Desai 2021).
- All are obtained by imposing g_tot = √($a_0$ g_bar), that is, the MOND form for very low accelerations.
- And the *mass deficit* of MOND in clusters has long been known: even with MOND, mass is missing.
- **Modifications of MOND that depend on the potential** in clusters: EMOND (Zhao and Famaey 2012; Hodson and Zhao 2017).
- **The splashback radius** of clusters: about 0.8-1.5 R200m, that is, roughly 1.5-2 R200c for massive clusters at low redshift, that is, relatively close to us, in space and in time (More, Diemer and Kravtsov 2015; Bianconi et al. 2021).

## 2. The reading used

**Two ratios.**

- L = R_s/r. R_s is the *Schwarzschild radius* of the mass: the radius that a black hole with that mass would have. L compares the mass within r with the maximum that that radius can contain. The framework calls it "stacked cubes".
- S = r/R_H. R_H = $c/H_0$ is the *Hubble radius*: the distance at which, by Hubble's law, galaxies recede at the speed of light. S compares the space up to r with the Hubble radius. The framework calls it "cubes side by side".
- The two ratios are equal at the radius r_t = √(R_s R_H).
- For g_N = GM/r², that is, Newton's acceleration, L/S = g_N/($cH_0$/2) holds exactly.

**The form of the RAR.**

- In the author's book, NUMERI, the ratio g_dark/g_bar is the sum of *reading steps* 1/(e^√(L/S) − 1). g_dark is the dark part, g_bar the baryonic one.
- This sum has the same form as the RAR formula.
- The step factor is a choice. The form of the RAR is taken from the data.
- Identifying the *two-end threshold* with the parameter $a_0$ of the formula is a further choice. Other definitions of the observed onset give other numbers. For example, if the onset is where the dark part equals the baryonic one, one gets g_bar = $(\ln 2)^2\, a_0$.

**The dictionary of the two mirrors.** It links the visible sector and the hidden one with two maps.

- *Map G* is the geometry of the reading between the two sectors: √3 per direction. Lengths are in the ratio √3, surfaces 3, volumes 3√3. Our own mass is not rescaled.
- *Map A* of MM 2.44 concerns the internal units of the hidden sector (A ≳ 1000, that is, at least about a thousand).
- If the two maps turned out to be the same conversion of lengths, this dictionary would fall.

## 3. The galaxy scale as $cH_0/6$

**The numbers.** ($cH_0$/2)/3 is:

- $1.091\cdot10^{-10}$ with $H_0$ = 67.4: −0.041 dex from 1.20;
- $1.182\cdot10^{-10}$ with $H_0$ = 73.0: −0.007 dex.

Both lie inside the *systematic band* of SPARC, $0.96\text{-}1.44\cdot10^{-10}$. SPARC is a catalogue of disc galaxies whose rotation is well known.

**How it was found.** The factor was counted as "two directions" knowing that it gives the right number.

- One looks at the family ($cH_0$/2)/3^(p/2). Only p = 2 falls within 0.1 dex of the measured value.
- But the step between one member of the family and the next is 0.239 dex, wider than that window. So at most one member can fall in it. This uniqueness is not evidence.
- The interest of the reading is coherence: the same rule on three directions is the 3√3 of MM 2.44.
- The coincidence with Verlinde concerns only the coefficient, the 6.

**Coherence checks.** They are approximate estimates.

- **Near the Sun.** One chooses the baryonic speed that gives a rotation of 229 km $s^{-1}$ for each scale: the rotation is imposed, not tested. Then the vertical *surface density* within 1.1 kpc is 63.9 $M_\odot\ pc^{-2}$ with $cH_0/6$ and 107.4 with $cH_0$/2. The measurement is 68 ± 4. $M_\odot$ is the mass of the Sun; pc is the parsec, a little more than three light years; kpc means a thousand parsecs.
- **The baryon fraction of clusters.** With 3√3 it is 1/(1 + 3√3) = 0.161. It does not contain $a_0$, so it stays unchanged by construction. In an example cluster, with $10^{14}\ M_\odot$ of baryons within 1 Mpc, the RAR mass drops by 3.9 %. So the known deficit grows slightly.
- **The evolution in time, $a_0 \propto H(z)$.** The four bins of MUSE-DARK III, an observation of distant galaxies, are used. The errors are made symmetric and there are no covariances.
- With free amplitude, the shape is compatible if the error bars are at 1σ ($\chi^2$ = 5.48 for 3 degrees of freedom, p = 0.14). It is not if the bars are at 95 % ($\chi^2$ = 21.04, p ≈ $10^{-4}$).
- With each of the three fixed normalisations considered ($cH_0/6$ with two values of $H_0$, and 1.20) the $\chi^2$ are high. The bins lie on average 16-27 % higher.

**Not established.**

- That the field spreads exactly over two directions between the mirrors.
- The direction of the factor.
- The identification with $a_0$.
- $cH_0/6$ and $cH_0/2\pi$ differ by 4.7 %. Today they cannot be separated.

## 4. The threshold and the readings that fall

**The threshold written with counts.**

- One defines a *capacity* $C(r) = c^2r/(2Gm)$, the *stacked cells* $N = M/m$ and $U = M_U/m$. Here m is the mass of a cell.
- The onset is where $C(r)^2 = N \cdot U$. It gives $r_{on}^2 = R_s(M) \cdot R_s(M_U)$.
- With $g = GM/r^2$ (spherical-equivalent mass) the threshold is $g = c^4/(4G \cdot M_U) = c^2/(2R_s(M_U))$. The cell m cancels.
- With the true mass of a disc a geometric factor can reappear.
- Formally this is the *surface gravity* of a Schwarzschild mass $M_U$. It is an identity, not an identification.

**Which M_U.**

- If $M_U = M_H = c^2R_H/(2G)$, that is, the mass-energy of a critical-density sphere of radius $R_H$, taken as the limit of the reading, one gets $cH_0/2$.
- To reproduce $a_0$ one needs $M_U = 2.728\ M_H$. It is one quarter of Milgrom's "MOND mass", $c^4/(Ga_0)$. It is a required value, not a prediction. And it does not establish a cosmological origin.
- Other boundaries were examined: matter only, event horizon, particle horizon, de Sitter radius. They give from 0.3 to 32 $M_H$. None is within 0.1 dex of 2.73.

**Premise 43 and the readings that fall.** Premise 43 tied the onset of the dark part to the centre of each galaxy.

- The registered readings are nine: the two-end reference and eight others.
- The galaxies are 148 from SPARC. Criteria: quality $Q \leq 2$, inclination $\geq 30°$, at least five points, increasing radii, positive baryonic acceleration. The central density comes from the first point; the masses of the components as $v^2r/G$.
- **The results:**
- the reference gives a single scale that is too high by 0.44 dex;
- the readings with the central density give scales from 70 to 2000 times too high;
- the readings with the central potential or with the constituents' readings (T1, T4, T5, T7) move the onset to lower accelerations, but scatter by 0.79-0.99 dex from galaxy to galaxy;
- the "mcm" built from the masses of the constituents of a galaxy (maximum or sum, in the conventions of readings T8a and T8b) gives thresholds too high by 13.3 and 13.2 dex. "mcm" is the abbreviation of *minimo comune multiplo*, the least common multiple, a word of the framework, like the MCD (the greatest common divisor);
- the scatter of the eight readings ranges from 0.26 to 0.99 dex.
- **None of the readings tried passes both registered criteria.** These readings fall. The outcome hits the readings that tie the onset to the centre of the galaxy, not the MM framework.
- **An observed fact.** In the 63 galaxies in which the onset can be located, its radius, measured in disc lengths, grows with the central density of the disc (Spearman +0.52). This is what a single acceleration scale implies.

## 5. Twelve X-COP clusters

**Data and model.**

- The total masses are hydrostatic, in two versions: the NFW model and a "forward" reconstruction. Both assume hydrostatic equilibrium and spherical symmetry.
- There are the gas masses and the stellar profiles for 7 clusters. For the other 5 the median star/gas ratio of those 7 is used.
- Two profiles end before 1.2 R500: the gas of A3266 at 1.12 and the stars of A2319 at 1.06 R500. Beyond, the last value is held. The preference for A/r persists also in the two NFW fits truncated to the available coverage. The extended stellar profile of A2319 also enters the median star/gas ratio used for five clusters.
- The stellar profiles may be *projected*, that is, seen on the sky: about 2 % on a (≈ 2 %), from an earlier check not recomputed by the module.

**The registered test: no indication.**

- One scale per cluster at R500 is compared with c200, the *concentration*, and with the *central entropy* of the gas, measured at 0.03 R500. For A2255 and A2319 the entropy profile starts beyond that radius, and the innermost bin is used.
- The Spearman values are −0.09 and +0.02; +0.16 without those two clusters. No indication.
- At 0.5 R500 the concentration is linked to the scale, but with the sign opposite to the registered one ($p = 0.006$). It comes from an earlier check, not recomputed by the module.

**Observed afterwards, declared as such.** The scale is computed with $a = g_{bar}/[\ln(1 - f_b)]^2$, where $f_b = g_{bar}/g_{tot}$ is the baryonic fraction of the acceleration.

- In all twelve clusters the scale decreases with radius.
- The median slope is −0.97, with a range from −1.44 to −0.61, between 0.2 and 1.2 R500.

**A/r against a constant scale.** Each cluster is given an amplitude. The deviation is measured with the *RMS*, the root of the mean of the squares of the residuals in log g_tot, over 21 interpolated radii. They are not independent measurements.

- A/r does better in 12 clusters out of 12 (12/12).
- Mean RMS: 0.023 against 0.105 dex with the NFW mass; 0.040 against 0.121 dex with the forward mass.
- But $r \cdot a(r)$ is constant only approximately. Within a cluster it varies by up to ×2.2 (NFW) and ×6.5 (forward).

**The MM reading.**

- In galaxies $S = r/R_H$ grows point by point.
- In clusters S stays about fixed: $S = R_{fix}/R_H$. Then $L/S \propto G \cdot M_{bar}(<r)/r$. It is a quantity with the dimensions of a potential ($v_{bar}^2$), not the full gravitational potential of an extended body.
- From this comes $R_{fix} = 2A/(cH_0)$. The median is 0.92 R500 (NFW) or 0.98 R500 (forward).
- With the factor 1/3 of section 3 it becomes $6A/(cH_0)$: median 1.82 R200c (NFW) or 1.83 R200c (forward). It is a scale compatible with the splashback radii of massive clusters.
- **Be careful:** these are medians of a conversion of A, not measured boundaries. The splashback radii of these clusters are not measured here.

**The sensitivity to the total mass (added in version 2.46.1).**

- For small baryon fractions $a \approx g_{tot}^2/g_{bar}$. At fixed radius and baryonic mass, $d \ln a / d \ln M_{tot} = 2.06$-$2.27$ over the baryon fractions of the fitted X-COP nodes. In words: if the total mass grows a little, the scale grows about twice as much, in proportion.
- With fixed radii, baryonic profiles and reference R500:
- increasing the X-COP total masses by 10 % increases the fitted R_fix by about 23 %;
- decreasing them by 10 % reduces it by about 20 %.
- A1795 and A2029 were reconstructed with the gas and NFW mass models of Vikhlinin et al. (2006), using the median X-COP star/gas ratio. They give $3R_{fix}/R200c = 3.3$ and 3.0 instead of 1.8 and 2.2.
- It is a comparison between reconstructions: the gas model and the fitted range differ too.
- Vikhlinin's total mass (V06) for A1795 is 30-37 % higher at the three radii checked: 0.3, 0.6 and 1.0 Mpc.
- **What follows from this.** The radius of the fixed end depends strongly on the mass profiles adopted. Its closeness to the splashback scales therefore depends on the calibration.

**The ordinary explanation.** For small baryon fractions $d \ln a / d \ln r \approx m - 2 - \beta$.

- $m = d \ln M_{tot}/d \ln r$ is the slope of the total mass; β (read *beta*) $= d \ln f_b/d \ln r$ is the slope of the baryon fraction.
- NFW-type masses ($m \approx 1.2$) with rising baryon fractions ($\beta \approx 0.2$) give about $-1$ ($\approx -1$).
- The data do not distinguish this explanation from the MM reading.

**Other samples.** If the profile decreases with radius, the constant scale derived from it is higher where the inner radii count more. This may explain part of the difference among CLASH, X-COP and groups. Not all of it: A differs too.

## 6. The CLASH cores

**Premise 47.** In a cluster the galaxies are far apart even if in projection they look close. In a galaxy the stars are close. The type of *fractionation*, that is, of division into parts, seen from the observer changes the observed behaviour.

**The registered test.** Inside the brightest galaxies of the clusters, called BCGs, the scale should read as in galaxies.

**Registered outcome: no indication.**

- In 10 cores out of 13 the central values of the baryonic mass exceed the lensing mass. It is a tension between estimates, not a measurement of zero dark matter.
- The other three give a median of $1.67 \cdot 10^{-10}$, just outside the registered band.

**Exploratory, added afterwards.** The weights are simplified: the lensing error and 10 % on the stars. The errors of the gas and their propagation through the non-linear prediction are not included. The ratio between the predicted mass and the lensing mass is:

- 2.0-6.4 (median 3.7) for the constant scale of the clusters;
- 5.0-15.3 (median 10.4) for A/r with the X-COP amplitude;
- 0.81-2.42 (median 1.53) for the galaxy law;
- 0.66-1.98 (median 1.20) with no dark part.

Among the four prescriptions, the one with no dark part has the lowest score: $\chi^2$ 27, against 62, 1112 and 11,709 for the other three.

- With these masses and prescriptions the extensions of the cluster laws overestimate the cores.
- The galaxy scale is not singled out by the data.
- Tian et al. report instead that the BCGs lie on the cluster relation, in their combined fit. The difference with this core-by-core comparison is not resolved here: it could depend on the masses, on the function or on the way the accelerations are built.

## 7. What is not claimed

- **No derivation.** $a_0 = cH_0/6$, the two directions, the fixed distal end and its position are readings of measured numbers, found after seeing them.
- **No distinction** from the ordinary explanations of the cluster profiles, and **no observational confirmation** of MM.
- **No preference** for $cH_0/6$ over $cH_0/2\pi$ or over a constant $1.20 \cdot 10^{-10}$.
- **The mass deficit of clusters** of RAR-type laws is not solved. The scale of the cores is not measured.

## 8. How this reading can turn out wrong

Four tests are registered, written before the analysis of the samples they name. Each has three decision zones: agreement, an intermediate zone and a zone in which the reading falls. The four tests are:

- the slopes of the clusters in new samples;
- the fixed distal end against the measured splashback radii (test R2);
- $a_0/(cH_0/6)$ with independent calibrations;
- the cores at their measured accelerations.

**The rules on changes.**

- The published version of this record fixes the protocol before the analysis of the samples it names. Later changes will be registered separately.
- In version 2.46.1 the registered protocol of R2 and the decision criteria are unchanged.
- Module 06 documents how much R_fix depends on the calibration of the total mass adopted. R2 will be evaluated according to the published criteria. Any other sensitivity analysis will be reported separately.
- Any revised protocol will be dated and registered before the analysis to which it applies.

## How the work was done

This section gathers the procedure. In the archived text it corresponds to the heading and to sections 9, 10 and 11.

**Who did what.** The framework and its premises are the author's. The author is not a professional physicist. Translations, code and text were written with an AI assistant (Claude). The checks by a second AI assistant (Codex, "Astra") are

documented in the reports kept in `documenti/rapporti_astra/`, in Italian. There was no human *referee*, that is, an expert who checks a work before publication. The new numbers of the second assistant were recomputed with code.

**The registrations.** For each step the protocols and the local registrations are kept: a SHA-256 fingerprint and the local time, not an independent timestamp. A SHA-256 *fingerprint* is a code computed from the content of a file: if the file changes even slightly, the fingerprint changes.

**The working names.** In the working records this deposit collects premises 43-47 (folders 30-34 of the working tree); some file names carry those numbers. They belong to this record. The check added in version 2.46.1 comes from premise 48 (folder 36).

**The licences.** Original text and materials: CC BY 4.0. Original code: MIT licence. Third-party data are not covered by these licences and keep the conditions of their sources (`fonti/SOURCES.md`).

**The files cited in the sections.**

- Section 2: the dictionary of the two mirrors is `registrazioni/premessa_45/DIZIONARIO_DUE_SPECCHI.md`.
- Section 3: `codice/03_two_mirror_scale.py`.
- Section 4: `codice/01_threshold_and_mcm.py` and `codice/02_sparc_central_readings.py`.
- Section 5: `codice/04_xcop_clusters.py`; the sensitivity to the total mass is `codice/06_mass_sensitivity.py`.
- Section 6: `codice/05_bcg_cores.py`.
- Section 8: the four tests are in `registrazioni/REGISTRATION_2.46_EN.md`.

**How to redo the calculations.** One runs `python scarica_dati.py` once: it downloads SPARC from CDS and the X-COP release, and checks the SHA-256. Then one runs `python riproduci.py`: 6 modules, 63 checks of the main numerical results. They are reproduction and consistency checks, not physical confirmations. The download path is not among them.

**What is in the package.**

- `scarica_dati.py` checks the archive, every file and the completeness against the manifest. `riproduci.py` checks the local data against the manifest, then runs the modules. `codice/` contains six modules and `comune.py`. `risultati/` contains the reference JSON and logs per module, and `00_summary.json` with the package versions. Python 3.11+ with numpy, scipy, sympy and astropy is needed.
- `dati/`: the MUSE-DARK III bins, the CLASH table and, for module 06, the Vikhlinin et al. (2006) parameters of A1795 and A2029 (transcriptions of published numbers), and `MANIFEST_SHA256.txt` of the third-party files downloaded by `scarica_dati.py`.
- `registrazioni/premessa_43-premessa_48`: the author's words (in Italian), translations and criteria registered before each calculation, the original scripts and logs (with the paths of the author's machine) and the `REGISTRO.txt` of each folder; `REGISTRATION_2.46_EN.md`.
- The scripts of `premessa_48` are records of the work; they are not part of the 63 checks. They also read the arXiv sources of Vikhlinin et al. 2006, Gastaldello et al. 2007 and Sadhu and Tian 2024, which are not redistributed: `registrazioni/premessa_48/HOW_TO_RERUN.md` says how to download and check them.
- `documenti/note_di_verifica` (outcome notes, in Italian; corrections in separate files), `documenti/rapporti_astra` (reports of the second assistant, in Italian), `LEGGIMI_IT.md`, `WHERE_THIS_COMES_FROM.md`, `fonti/SOURCES.md`.

**The changes of version 2.46.1** (13 October 2026): module 06 and the paragraph "Sensitivity to the total mass" of section 5; the note on R2 in section 8; `dati/vikhlinin2006`; the working records of premise 48 with their notes and their reports. Modules 01-05 and their reference results are those of 2.46.

**Related records.** MM 2.44 (DOI 10.5281/zenodo.23329080: $3\sqrt{3}$, $a_0 \propto H(z)$), MM 2.42.1 (DOI 10.5281/zenodo.23329094).

**This version.** This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

## Glossary

**The words of the framework (MM)**

- **Fixed distal end.** The MM reading of the A/r law in clusters: S = r/R_H stays about fixed.
- **Hidden sector, mirror.** In MM 2.44, a matter that we do not see, read as dark matter, next to the visible sector.
- **Premise.** A starting sentence of the author, a fixed point like a postulate. The premises are numbered.
- **Reading.** An interpretation, written so that it can be put to the test.
- **Registered.** Written and fixed with its date before the calculations.
- **Two-end reading.** The MM reading that gives the scale $cH_0/2$.

**The words of physics**

- **$a_0$.** The acceleration scale of the RAR: around $a_0$ and below, the dark part becomes important.
- **Acceleration.** How much the velocity changes every second, in size or in direction.
- **Baryonic mass.** The mass of the matter made of atoms: stars and gas.
- **$cH_0$.** The product of the speed of light and the Hubble constant: an acceleration.
- **$\chi^2$ (chi-squared).** A number that measures how far a model departs from the data.
- **dex.** A way of measuring ratios in powers of ten: 1 dex is a factor of ten.
- **Galaxy cluster.** A very large group of galaxies bound by gravity, with a very hot gas between them.
- **Gravitational lensing.** The bending of light caused by mass; it makes it possible to weigh also the invisible mass.
- **Hubble radius (R_H).** $c/H_0$: the distance at which, by Hubble's law, galaxies recede at the speed of light.
- **Hydrostatic mass.** The mass derived from the hot gas in equilibrium between pressure and gravity.
- **MOND.** Milgrom's idea: below $a_0$ gravity is stronger than Newton's.
- **NFW.** A mass profile typical of dark-matter haloes in simulations.
- **p-value.** The probability of finding a deviation this large by pure chance.
- **R500, R200c, R200m.** Reference radii of a cluster, defined by the mean density inside them.
- **RAR.** The radial acceleration relation of galaxies: g_obs = g_bar/(1 − e^(−√(g_bar/$a_0$))).
- **RMS.** The root of the mean of the squares of the deviations.
- **Schwarzschild radius (R_s).** The radius that a black hole with a given mass would have.
- **Spearman.** A coefficient between −1 and +1 that says how much two quantities grow together.
- **Splashback radius.** The point where the matter falling into a cluster turns back for the first time.

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